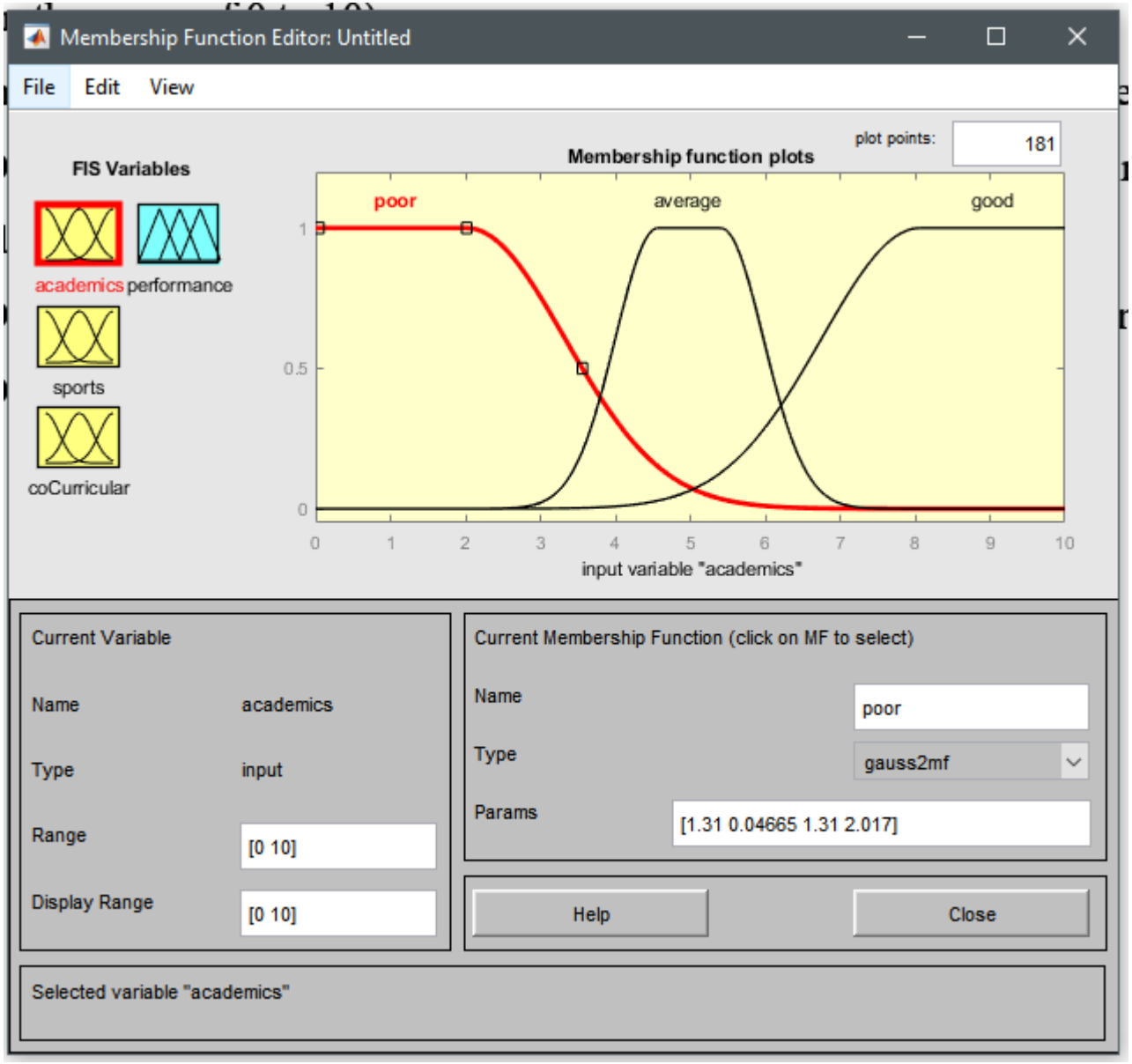
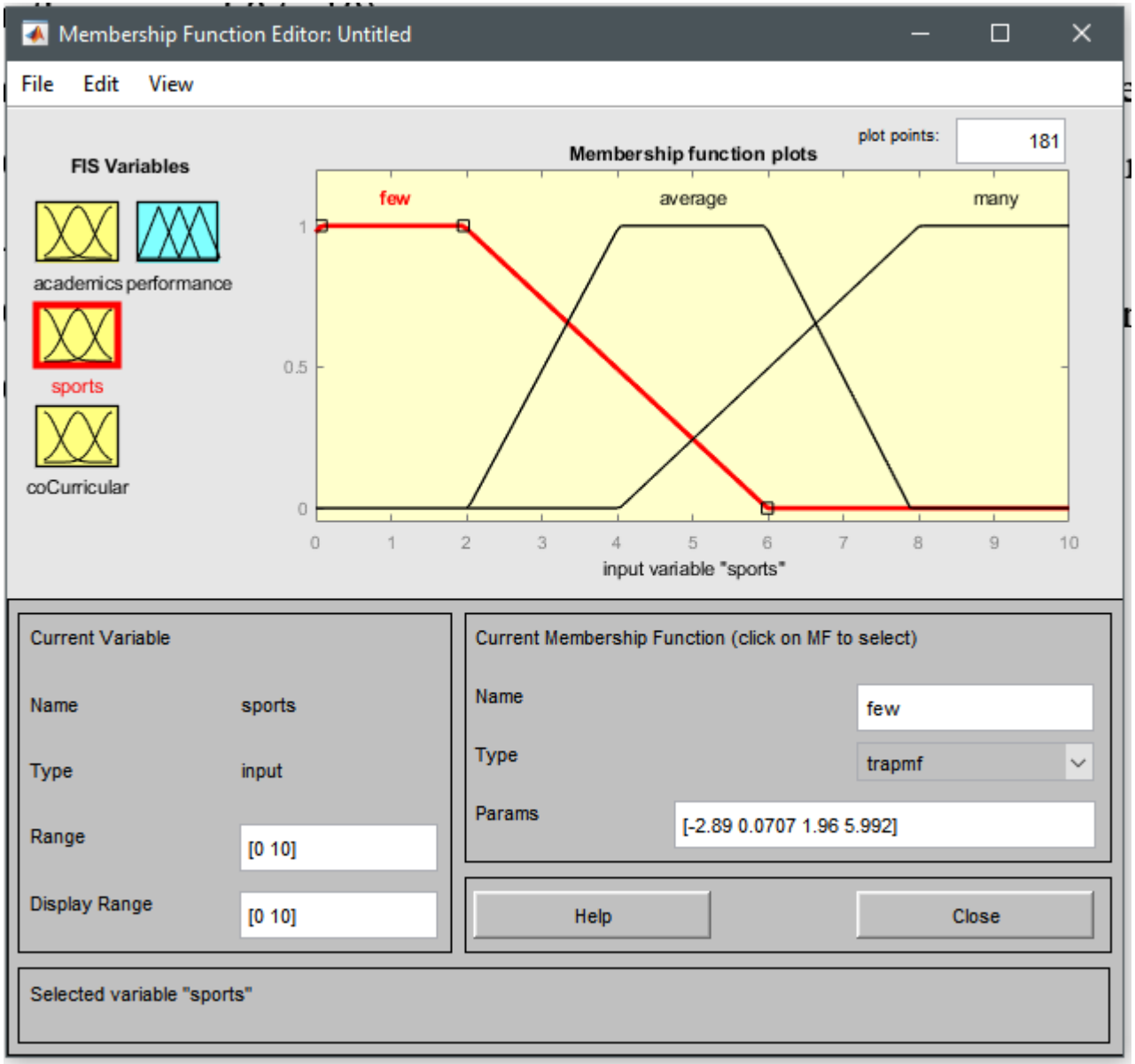
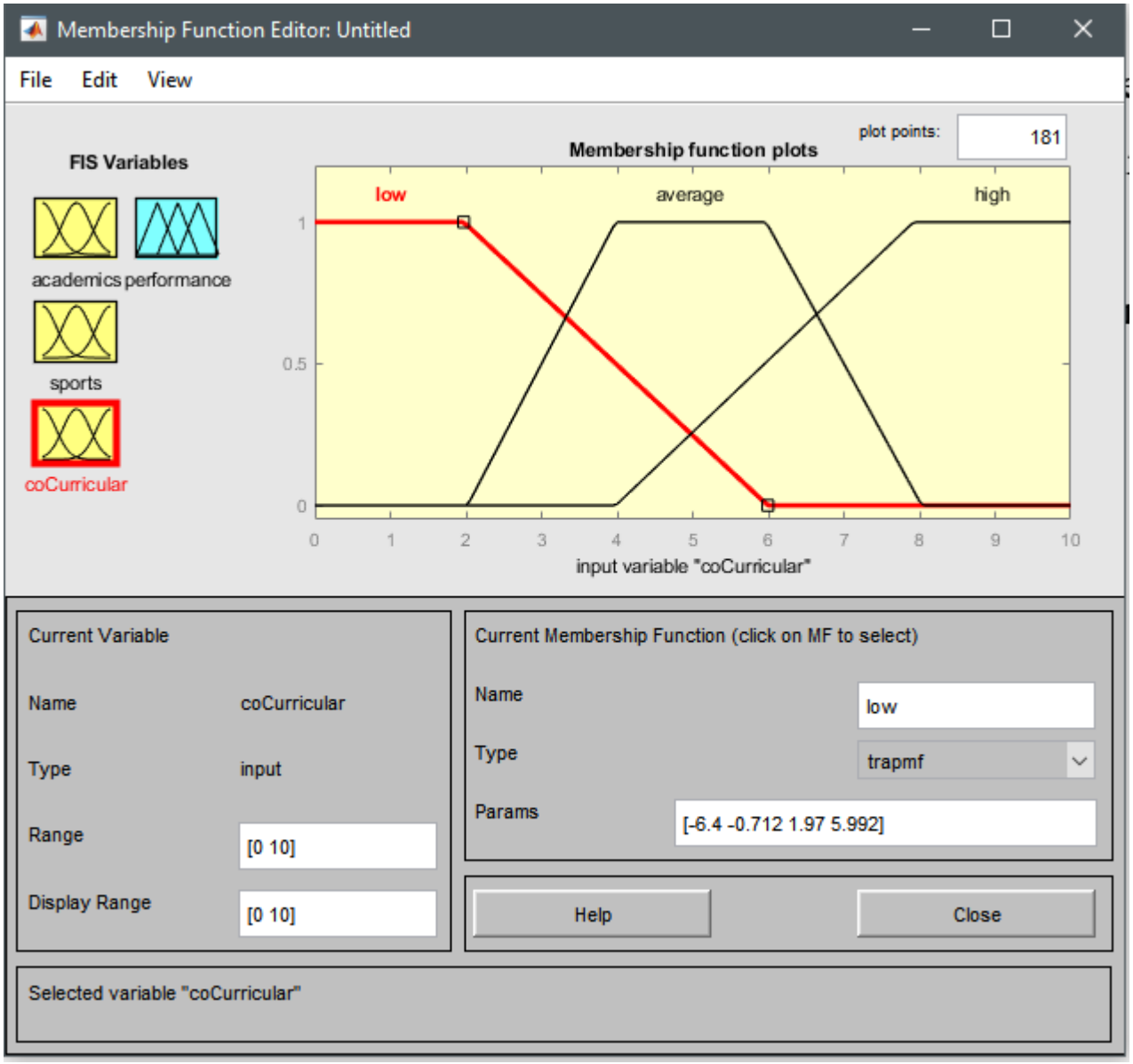
Artificial Intelligence – Assignment 4

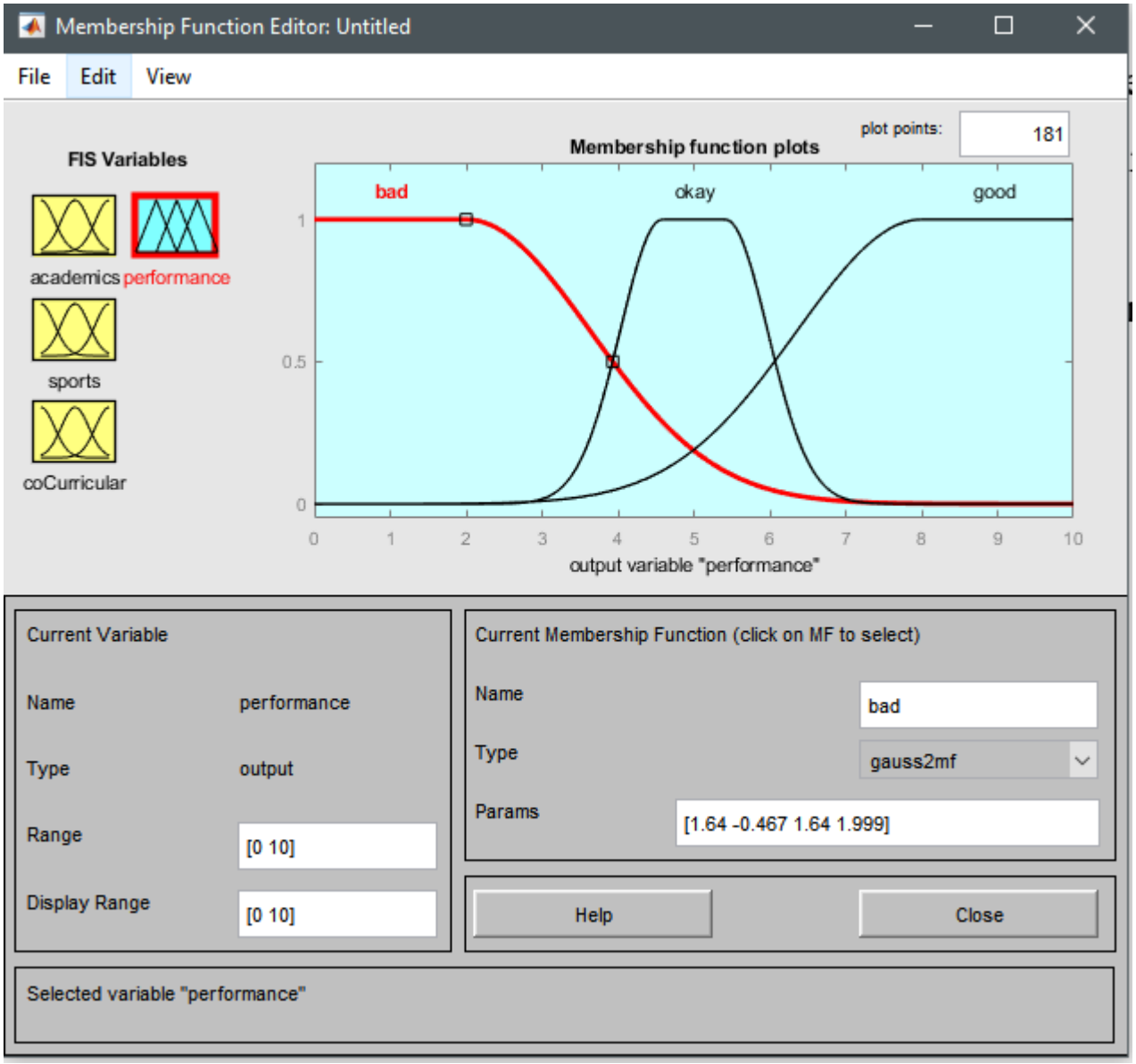
Abhyudaya Sharma – 1710110019

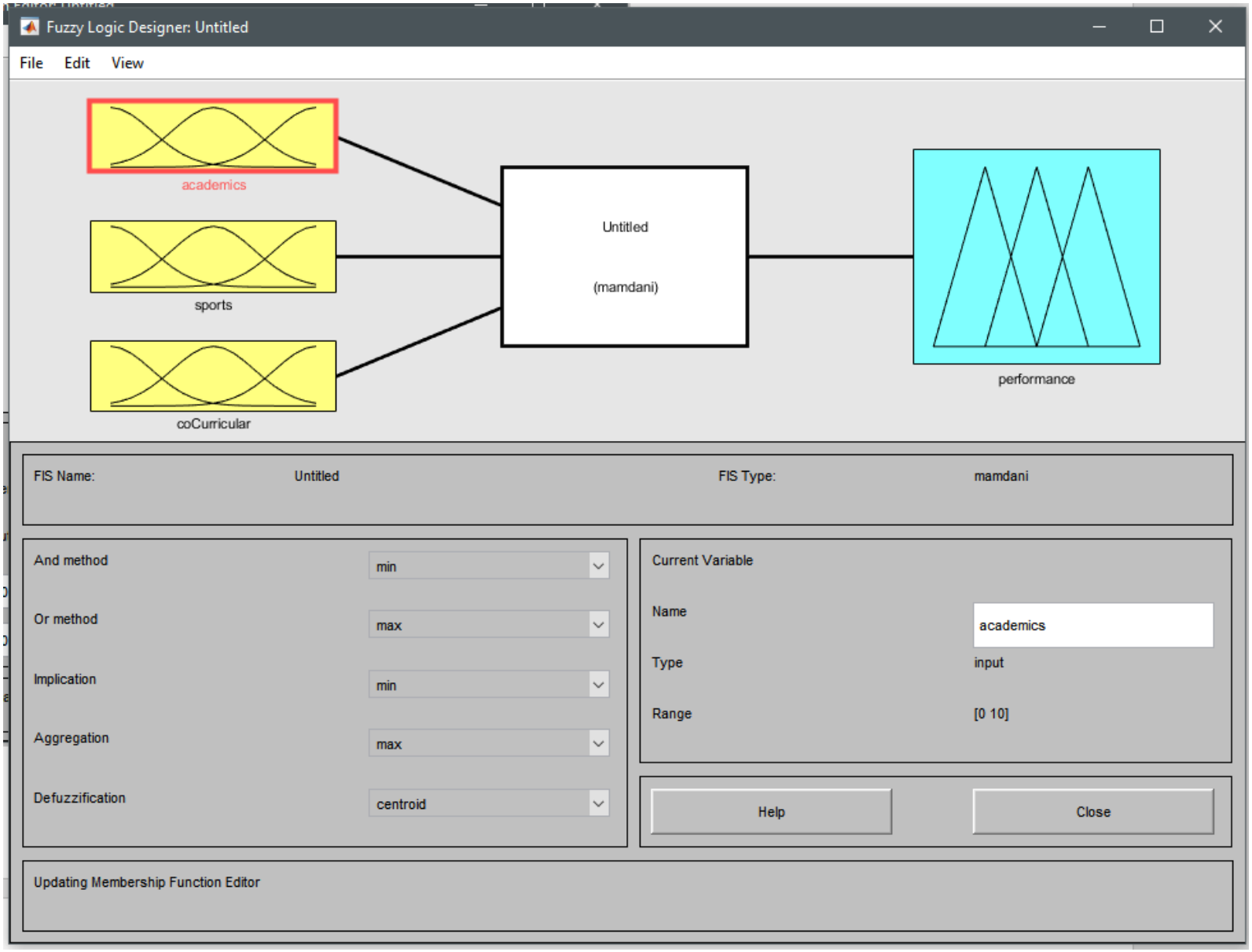
Kabir Kanha Arora - 1710110165

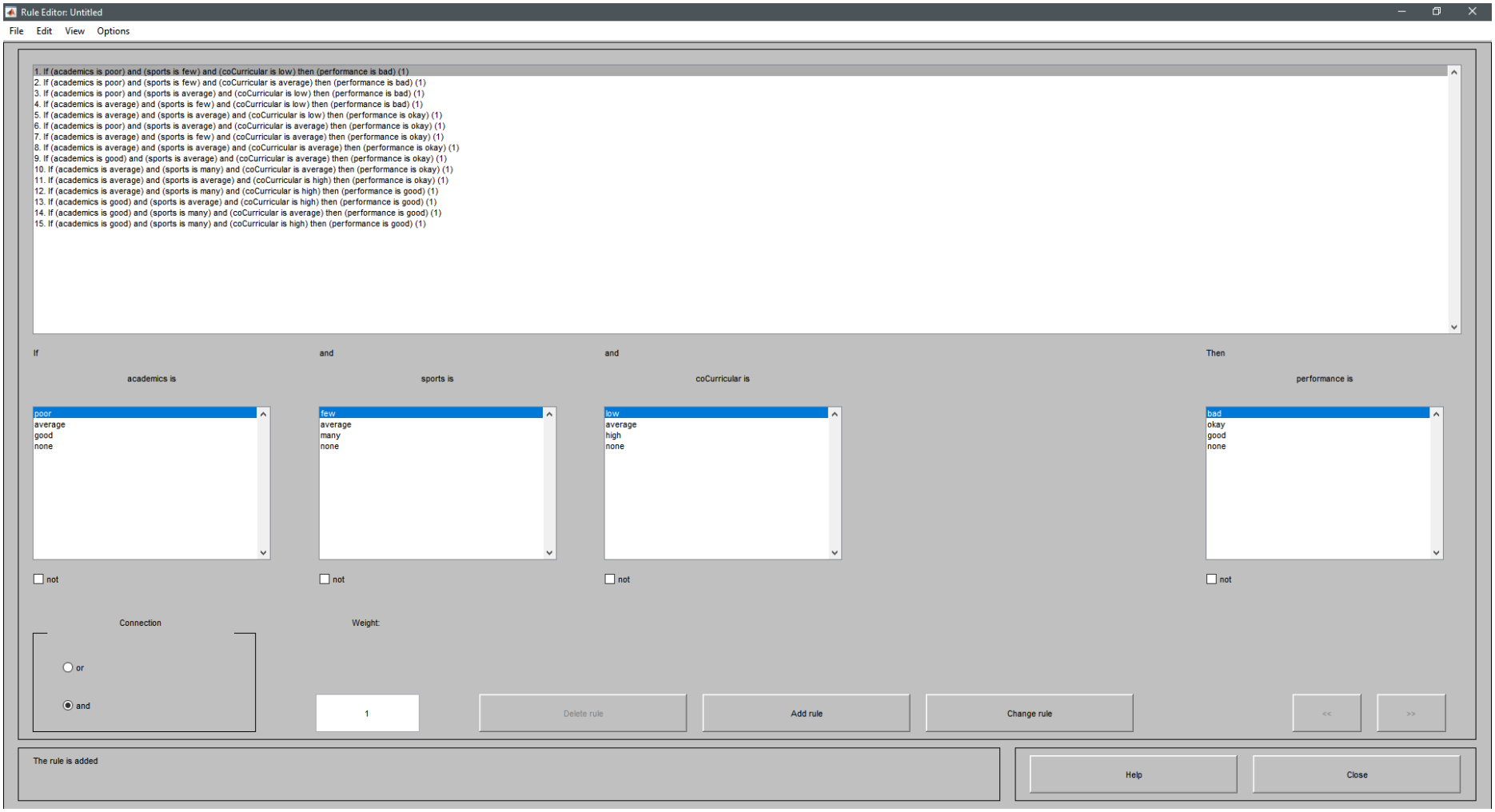

Membership Function Editor: Untitled
File
Edit
View
FIS Variables
academics
performance
sports
coCurricular
Membership function plots
plot points:
181
poor
average
good
1
0.5
0
0
1
2
3
4
5
6
7
8
9
10
input variable "academics"
Current Variable
Name
academics
Type
input
Range
[0 10]
Display Range
[0 10]
Current Membership Function (click on MF to select)
Name
poor
Type
gauss2mf
Params
[1.31 0.04665 1.31 2.017]
Help
Close
Selected variable "academics"

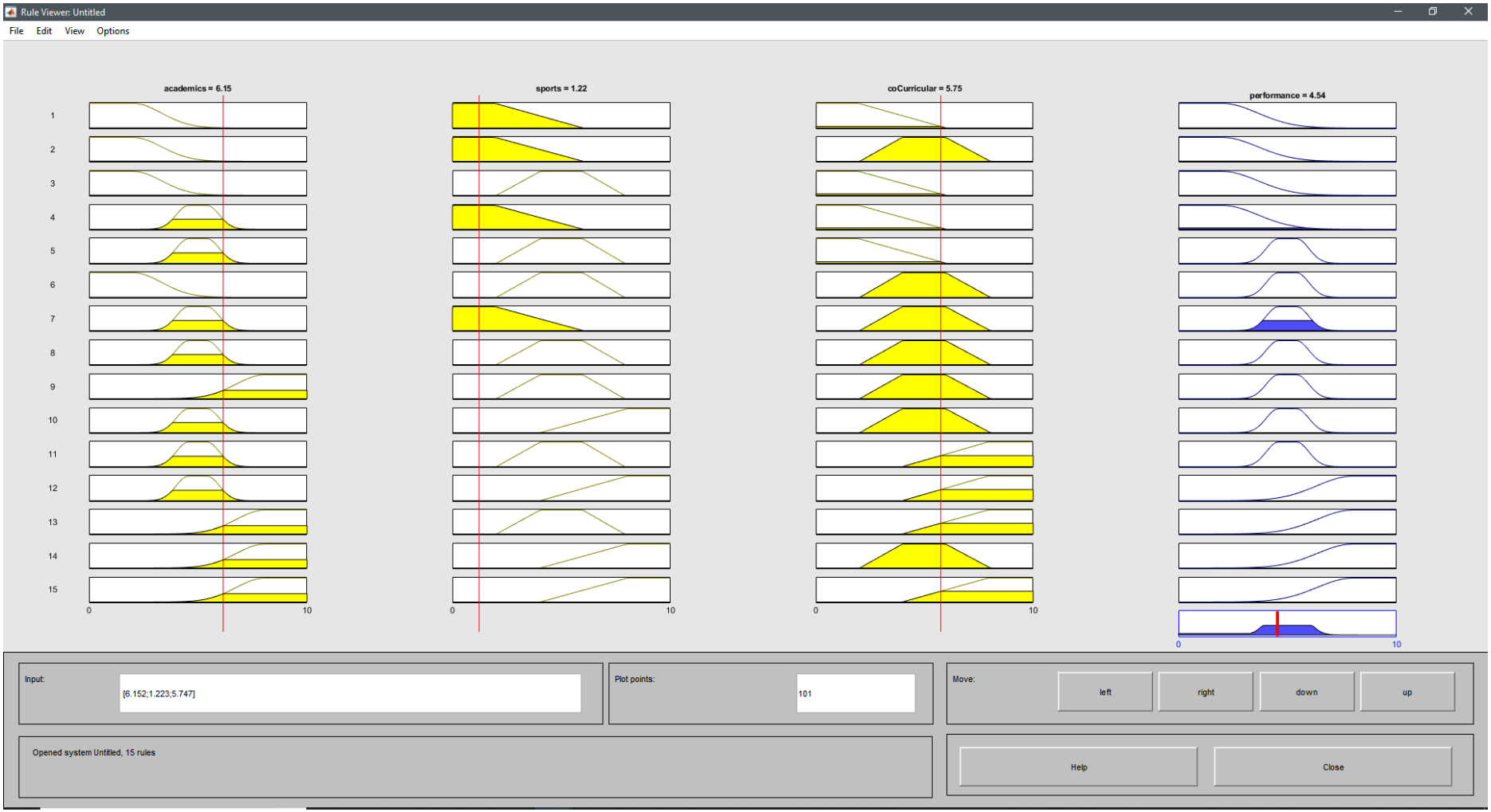

Membership Function Editor: Untitled
File
Edit
View
FIS Variables
academics
performance
sports
coCurricular
Membership function plots
plot points:
181
few
average
many
1
0.5
0
0
1
2
3
4
5
6
7
8
9
10
input variable "sports"
Current Variable
Name
sports
Type
input
Range
[0 10]
Display Range
[0 10]
Current Membership Function (click on MF to select)
Name
few
Type
trapmf
Params
[-2.89 0.0707 1.96 5.992]
Help
Close
Selected variable "sports"

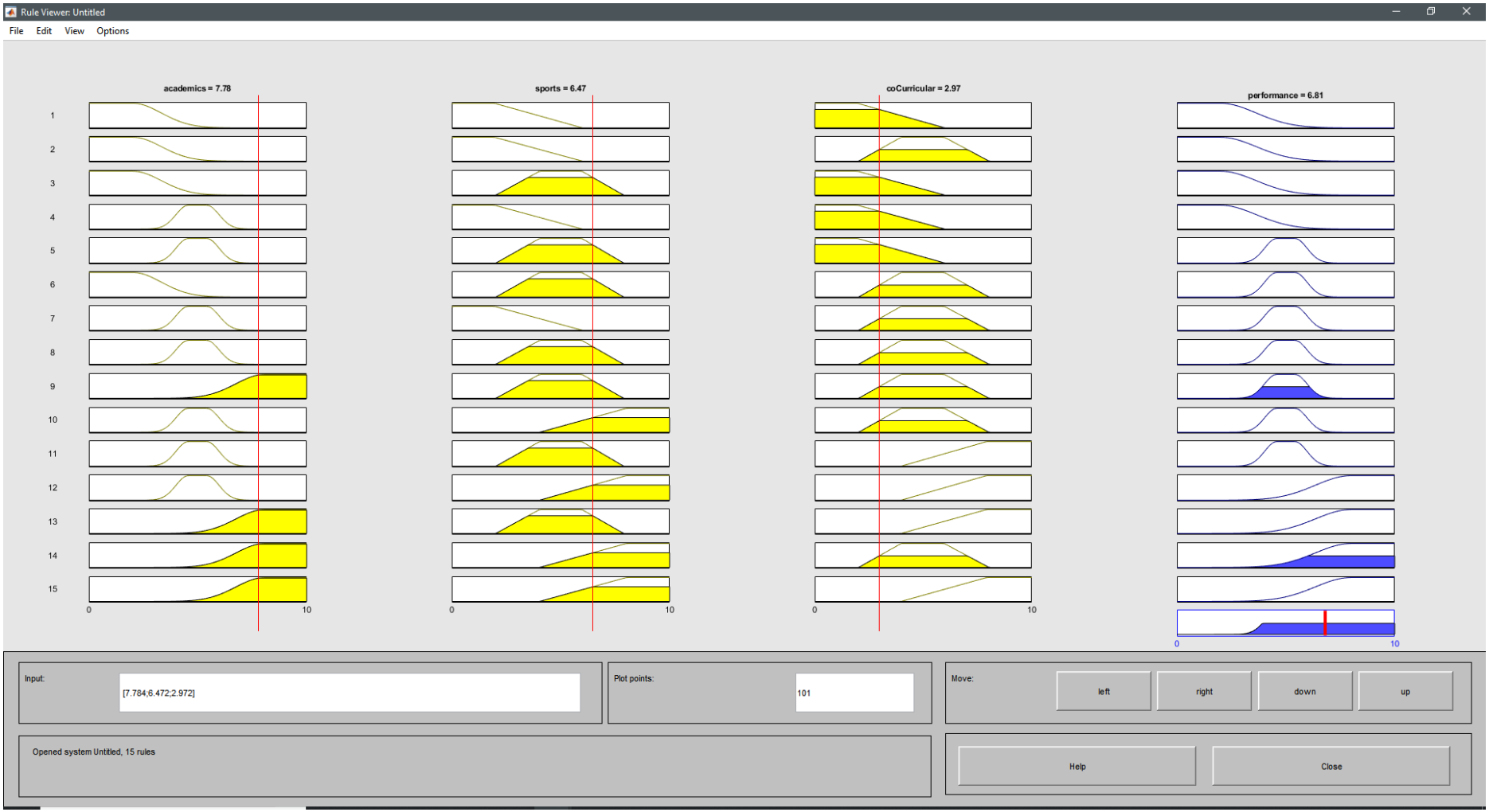

Membership Function Editor: Untitled
File
Edit
View
FIS Variables
academics
performance
sports
coCurricular
Membership function plots
plot points:
181
low
average
high
1
0.5
0
0
1
2
3
4
5
6
7
8
9
10
input variable "coCurricular"
Current Variable
Name
coCurricular
Type
input
Range
[0 10]
Display Range
[0 10]
Current Membership Function (click on MF to select)
Name
low
Type
trapmf
Params
[-6.4 -0.712 1.97 5.992]
Help
Close
Selected variable "coCurricular"

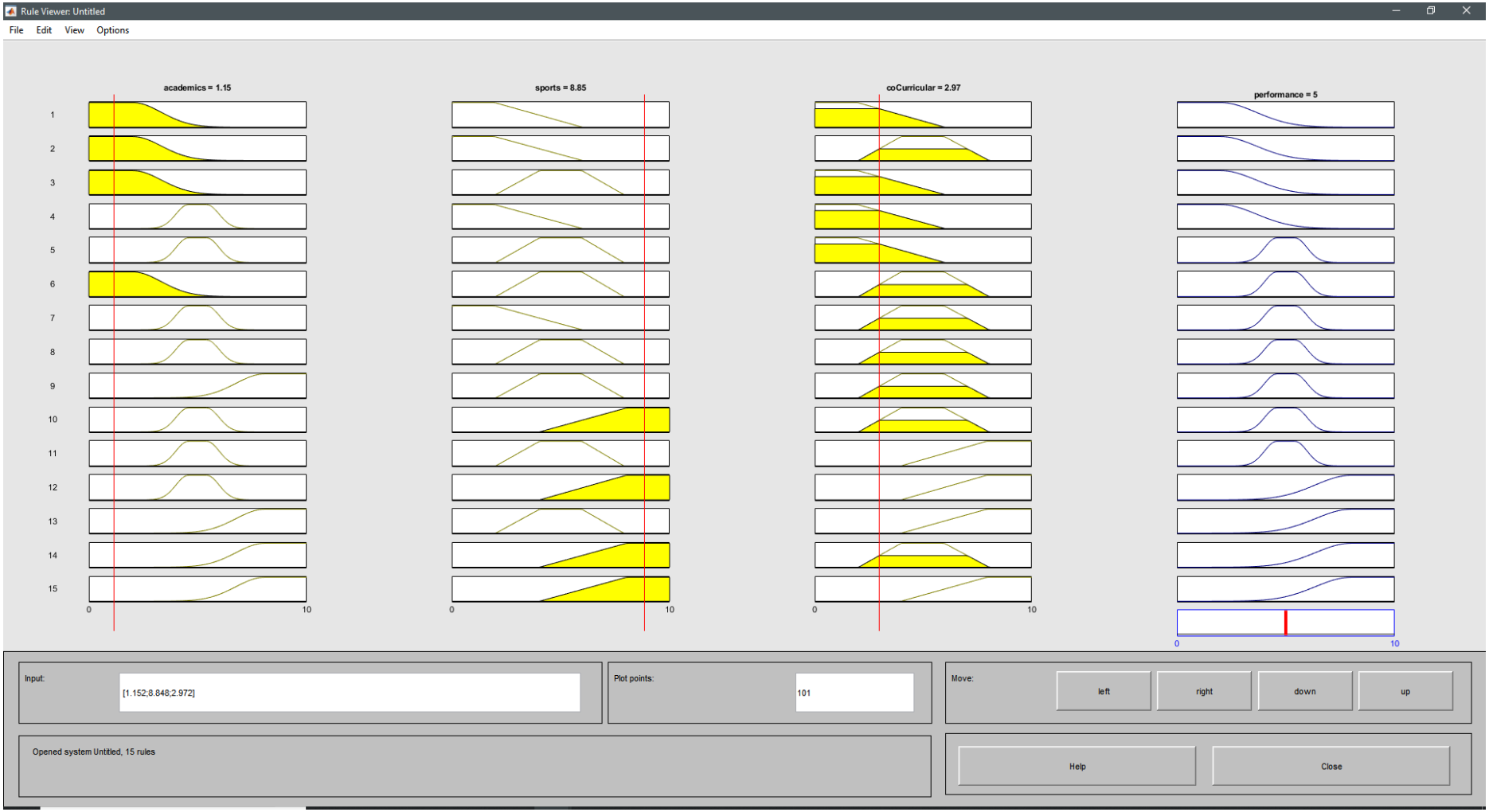

Membership Function Editor: Untitled
File
Edit
View
FIS Variables
academics
performance
sports
coCurricular
Membership function plots
plot points:
181
bad
okay
good
1
0.5
0
0
1
2
3
4
5
6
7
8
9
10
output variable "performance"
Current Variable
Name
performance
Type
output
Range
[0 10]
Display Range
[0 10]
Current Membership Function (click on MF to select)
Name
bad
Type
gauss2mf
Params
[1.64 -0.467 1.64 1.999]
Help
Close
Selected variable "performance"


Fuzzy Logic Designer: Untitled
File
Edit
View
academics
sports
coCurricular
Untitled
(mamdani)
performance
FIS Name:
Untitled
FIS Type:
mamdani
And method
min
Or method
max
Implication
min
Aggregation
max
Defuzzification
centroid
Current Variable
Name
academics
Type
input
Range
[0 10]
Help
Close
Updating Membership Function Editor


Rule Editor: Untitled
File
Edit
View
Options
1. If (academics is poor) and (sports is few) and (coCurricular is low) then (performance is bad) (1)
2. If (academics is poor) and (sports is few) and (coCurricular is average) then (performance is bad) (1)
3. If (academics is poor) and (sports is average) and (coCurricular is low) then (performance is bad) (1)
4. If (academics is average) and (sports is few) and (coCurricular is low) then (performance is bad) (1)
5. If (academics is average) and (sports is average) and (coCurricular is low) then (performance is okay) (1)
6. If (academics is poor) and (sports is average) and (coCurricular is average) then (performance is okay) (1)
7. If (academics is average) and (sports is few) and (coCurricular is average) then (performance is okay) (1)
8. If (academics is average) and (sports is average) and (coCurricular is average) then (performance is okay) (1)
9. If (academics is good) and (sports is average) and (coCurricular is average) then (performance is okay) (1)
10. If (academics is average) and (sports is many) and (coCurricular is average) then (performance is okay) (1)
11. If (academics is average) and (sports is average) and (coCurricular is high) then (performance is okay) (1)
12. If (academics is average) and (sports is many) and (coCurricular is high) then (performance is good) (1)
13. If (academics is good) and (sports is average) and (coCurricular is high) then (performance is good) (1)
14. If (academics is good) and (sports is many) and (coCurricular is average) then (performance is good) (1)
15. If (academics is good) and (sports is many) and (coCurricular is high) then (performance is good) (1)
If
academics is
poor
average
good
none
not
and
sports is
few
average
many
none
not
and
coCurricular is
low
average
high
none
not
Then
performance is
bad
okay
good
none
not
Connection
or
and
Weight:
1
Delete rule
Add rule
Change rule
<<
>>
The rule is added
Help
Close


Rule Viewer: Untitled
File
Edit
View
Options
academics = 6.15
sports = 1.22
coCurricular = 5.75
performance = 4.54
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
0
10
Input:
[6.152;1.223;5.747]
Plot points:
101
Move:
left
right
down
up
Opened system Untitled, 15 rules
Help
Close


Rule Viewer: Untitled
File
Edit
View
Options
academics = 7.78
sports = 6.47
coCurricular = 2.97
performance = 6.81
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
0
10
Input:
[7.784;6.472;2.972]
Plot points:
101
Move:
left
right
down
up
Opened system Untitled, 15 rules
Help
Close


Rule Viewer: Untitled
File
Edit
View
Options
academics = 1.15
sports = 8.85
coCurricular = 2.97
performance = 5
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
0
10
Input:
[1.152;8.848;2.972]
Plot points:
101
Move:
left
right
down
up
Opened system Untitled, 15 rules
Help
Close

Our choice of graphs

The graphs we chose for the linguistic variables ideally represent the situation we're trying to model. We chose gauss2mf for academic performance because the CGPA of the students is a number in the subset of the real number domain between 0 and 10. This function clearly distinguishes between the poor, average and good students. This allows us to easily model the situation with smooth movement from one category of academic performance to the next. For the sports performance and co-curricular activities, we ended up using trapmf. With the values falling off quickly because the number of certificates a student can get is an integer number. This allows us to rate the student on the number of certificates he or she gets without trying to approximate the values to a floating point number.

The results produce correct and expected values for the final performance value. This fuzzy logic enables us to grade students on not just one parameter. This is, therefore, a much better estimate of the capabilities of a student. We believe that a similar system should be enabled in our schools and universities to grade the overall development of students while still considering exceptional performance in one or more fields of study to have a positive impact on the calculated performance score.

We would like to conclude with the fact that this model can be much improved and has the potential to revolutionize how students are graded. We believe that further research should be carried out in this direction. After public discourse, agreed-upon ruleset and model should be made the standard at all schools and universities all around the globe.